Chapter P6: Matter – models and explanations

Overview

Richard Feynman said: "If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence passed on to the next generations of creatures, what statement would contain the most information in the fewest words? I believe it is the atomic hypothesis (or the atomic fact, or whatever you wish to call it) that all things are made of atoms—little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another. In that one sentence, you will see, there is an enormous amount of information about the world, if just a little imagination and thinking are applied." (*Six Easy Pieces*, p.4)

In this chapter the particle model described by Feynman is used to predict and explain some properties of matter. Topic P6.1 explores the relationship between energy and temperature and the ways in which energy transfer transforms matter. Topic P6.2 considers how the particle model explains the differences in densities between solids, liquids and gases and the effect of heating both in terms of temperature changes and changes of state. Topic P6.3 considers the behaviour of materials under stress and how the particle model can explain differences in behaviour. Topic 6.4 deals with pressure, and how it varies in fluids (liquids and gases), with the particle model used to explain the law discovered by Robert Boyle. Finally, in Topic P6.5 ideas about forces and also the particle model are considered in the context of planets, moons and satellites in their orbit, and the formation of the solar system, before briefly describing the early stage of the Universe and the Big Bang.

Learning about matter and the particle model before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- be able to use a particulate model of matter to explain states of matter and changes of state
- have investigated stretching and compressing materials and identifying those that obey Hooke's law
- be able to describe how the extension or compression of an elastic material changes as a force is applied, and make a link between the work done and energy transfer during compression or extension
- have investigated pressure in liquids and related this to floating and sinking
- be able to relate atmospheric pressure to the weight of air overhead.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

P6.1 How does energy transform matter?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
It took the insight of a number of eighteenth and nineteenth century scientists to appreciate that heat and work were two aspects of the same quantity, which we call energy. Careful experiments devised by Joule showed that equal amounts of mechanical work would always produce the same temperature rise. Energy can be supplied to raise the temperature of a substance by heating using a fuel, or an electric heater, or by doing work on the material. Mass – the amount of matter in an object – depends on its volume and the density of the material of which it consists. The temperature rise of an object when it is heated depends on its mass and the amount of energy supplied. Different substances store different amounts of energy per kilogram for each °C temperature rise – this is called the specific heat capacity of the material.	1. a) define density b) describe how to determine the densities of solid and liquid objects using measurements of length, mass and volume M1c, M5c *PAG1*
	2. recall and apply the relationship between density, mass and volume to changes where mass is conserved: $\text{density (kg/m}^3) = \dfrac{\text{mass (kg)}}{\text{volume (m}^3)}$ M1a, M1b, M1c, M3c
	3. describe the energy transfers involved when a system is changed by heating (in terms of temperature change and specific heat capacity)
	4. define the term specific heat capacity and distinguish between it and the term specific latent heat
	5. a) select and apply the relationship between change in internal energy of a material and its mass, specific heat capacity and temperature: change in internal energy (J) = mass (kg) × specific heat capacity (J/kg° C) × change in temperature (°C) M1a, M1c, M3d b) explain how to safely use apparatus to determine the specific heat capacity of materials *PAG5*

Linked learning opportunities

Specification links

- How much energy do we use? (P2.1)
- What determines the rate of energy transfer in a circuit? (P3.4)
- How can we describe motion in terms of energy transfers? (P4.4)

Practical work

- Devise a method to measure the density of irregular objects.
- Measure the specific heat capacity of a range of substances such as water, copper, aluminium.
- Measure the latent heat of fusion of a substance in the solid state and the latent heat of vaporisation of a substance in the liquid state.
- Show that the same amount of work always results in the same temperature rise.
- Collect data, plot and interpret graphs that show how the temperature of a substance changes when heated by a

P6.1 How does energy transform matter?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
When a substance in the solid state is heated its temperature rises until it reaches the melting point of the substance, but energy must continue to be supplied for the solid to melt. Its temperature does not change while it melts, and the change in density on melting is very small. Similarly as a substance in the liquid state is heated its temperature rises until it reaches boiling point; its temperature does not change, although energy continues to be supplied while it boils. The change in density on boiling is very great; a small volume of liquid produces a large volume of vapour. Different substances require different amounts of energy per kilogram to change the state of the substance – this is called the specific latent heat of the substance.	6. select and apply the relationship between energy needed to cause a change in state, specific latent heat and mass: energy to cause a change of state (J) = mass (kg) × specific latent heat (J/kg) M1a, M1c, M3d
	7. describe all the changes involved in the way energy is stored when a system changes, and the temperature rises, for example: a moving object hitting an obstacle, an object slowing down, water brought to a boil in an electric kettle
	8. make calculations of the energy transfers associated with changes in a system when the temperature changes, recalling or selecting the relevant equations for mechanical, electrical, and thermal processes M1a, M1c, M2a, M3b, M3c, M3d

Linked learning opportunities

Ideas about Science

- Describe and explain how careful experimental strategy can yield high quality data (IaS1).
- Describe and explain an example of how developing a new scientific explanation takes creative thinking (IaS3).

P6.2 How does the particle model explain the effects of heating?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
The particle model of matter describes the arrangements and behaviours of particles (atoms and molecules); it can be used to predict and explain the differences in properties between solids, liquids and gases. In this model: • All matter is made of very tiny particles. • There is no other matter except these particles (in particular, no matter between them). • Particles of any given substance are all the same. • Particles of different substances have different masses. • There are attractive forces between particles. These differ in strength from one substance to another. • In the solid state, the particles are close together and unable to move away from their neighbours. • In the liquid state, the particles are also close together, but can slide past each other. • In the gas state, the particles are further apart, and can move freely. The particle model is an example of how scientists use models as tools for explaining observed phenomena. The particle model can be used to describe and predict physical changes when matter is heated. • The particles are always moving: in the solid state, they are vibrating; in the liquid state, they are vibrating and jostling around; in the gas state, they are moving freely in random directions. • A substance in the gas state exerts pressure on its container because the momentum of the particles changes when they collide with walls of the container. • The hotter something is, the higher its temperature is and the faster its particles are vibrating or moving. Careful experimentation and mathematical analysis showed that the temperature	1. explain the differences in density between the different states of matter in terms of the arrangements of the atoms or molecules 2. use the particle model of matter to describe how mass is conserved, when substances melt, freeze, evaporate, condense or sublimate, but that these physical changes differ from chemical changes and the material recovers its original properties if the change is reversed 3. use the particle model to describe how heating a system will change the energy stored within the system and raise its temperature or produce changes of state 4. explain how the motion of the molecules in a gas is related both to its temperature and its pressure: hence explain the relationship between the temperature of a gas and its pressure at constant volume *qualitative only*	**Ideas about Science** • Use the particle model to explain familiar or unfamiliar phenomena and make predictions (IaS3).

P6.3 How does the particle model relate to material under stress?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
When more than one force is applied to a solid material it may be compressed, stretched or twisted. When the forces are removed it may return to its original shape or become permanently deformed. These effects can be explained using ideas about particles in the solid state. A substance in the solid state is a fixed shape due to the forces between the particles. Compressing or stretching the material changes the separation of the particles, and the forces between the particles. Elastic materials spring back to their original shape. If the forces are too large the material becomes plastic and is permanently distorted. For some materials, the extension is proportional to the applied force, but in other systems, such as rubber bands, the relationship is not linear, even though they are elastic. When work is done by a force to compress or stretch a spring or other simple system, energy is stored, this energy can be recovered when the force is removed.	1. explain, with examples, that to stretch, bend or compress an object, more than one force has to be applied
	2. describe **and use the particle model to explain** the difference between elastic and plastic deformation caused by stretching forces
	3. a) describe the relationship between force and extension for a spring and other simple systems b) describe how to measure and observe the effect of forces on the extension of a spring M2b, M2f *PAG2*
	4. describe the difference between the force-extension relationship for linear systems and for non-linear systems
	5. recall and apply the relationship between force, extension and spring constant for systems where the force-extension relationship is linear force exerted by a spring (N) = spring constant (N/m) × extension (m) M1c, M3b, M3c
	6. a) calculate the work done in stretching a spring or other simple system, by calculating the appropriate area on the force-extension graph M4f b) describe how to safely use apparatus to determine the work done in stretching a spring (PAG 2)
	7. select and apply the relationship between energy stored, spring constant and extension for a linear system: energy stored in a stretched spring (J) =

Linked learning opportunities

Practical work

- Investigate the force-extension properties of a variety of materials, identifying those that obey Hooke's law, those that behave elastically, and those that show plastic deformation.

P6.4 How does the particle model relate to pressures in fluids? (*separate science only*)

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*		*Linked learning opportunities*
An object immersed in a fluid (a liquid or a gas) experiences forces acting at right angles to all its surfaces due to the pressure of the fluid. The pressure of the fluid is due to collisions of the particles of the fluid with the surface of the object. The particles of gas in a container collide with the surfaces of the container, exerting a pressure. If the volume of the container is increased, the particles have further to travel between collisions and the pressure of the gas falls. When a gas is compressed the particles are much closer together and will collide with the walls of the container more frequently, exerting a greater outward pressure. The atmosphere of the Earth exerts a pressure perpendicular to the surface of any object in it, and this pressure is the same in all directions at a particular height. Atmospheric pressure decreases with height above the surface of the Earth. The pressure at a point in a fluid increases with depth, because it is caused by the gravitational force on the fluid above that point. A fluid with greater density will experience a greater gravitational force and so exert a greater pressure.	1.	recall that the pressure in fluids causes a force normal to any surface	Practical work • Investigate the relationship between density of an immersed object and density of the fluid and the net force on the object. • Devise an experiment to show that pressure in a fluid varies with depth. • Investigate the relationships between the pressure of a gas and its volume and its temperature.
	2.	recall and apply the relationship between the force, the pressure, and the area in contact: $\text{pressure (Pa)} = \dfrac{\text{force normal to a surface (N)}}{\text{area of that surface (m}^2\text{)}}$ M3b, M3c	
	3.	recall that gases can be compressed or expanded by pressure changes and that the pressure produces a net force at right angles to any surface	
	4.	use the particle model of matter to explain how increasing the volume in which a gas is contained, at constant temperature, can lead to a decrease in pressure	
	5.	select and apply the equation: pressure (Pa) × volume (m^3) = constant (for a given mass of gas at constant temperature) M1c, M3b, M3c, M3d	
	6.	describe a simple model of the Earth's atmosphere and of atmospheric pressure, and explain why atmospheric pressure varies with height above the surface	
	7.	**explain why pressure in a liquid varies with depth and density**	

P6.4 How does the particle model relate to pressures in fluids? (*separate science only*)

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
Because pressure increases with depth, the force on the lower surface of an immersed object will be greater than the force on the upper surface, resulting in a net force upwards. This explains why the apparent weight of an object immersed in a liquid is less than its weight in air. It was Dalton's careful study of the atmosphere and gases that led to him giving a quantitative significance to the atomic theory which provides the basis for the particle model of matter.	**8. select and apply the equation to calculate the differences in pressure at different depths in a liquid:** **pressure (Pa) = density (kg/m^3) × gravitational field strength (N/kg) × depth (m)** **M1c, M3c** **9. explain how the increase in pressure with depth in a fluid leads to an upwards force on a partially submerged object** **10. describe and explain the factors which influence whether a particular object will float or sink**	

P6.5 How can scientific models help us understand the Big Bang? (*separate science only*)

Teaching and learning narrative

The gravitational interaction between the planets and the Sun keeps the planets in (almost) circular orbits around the Sun, all in the same direction. Similarly, it is the gravitational interaction between a planet and its moons or artificial satellites that keeps them in orbit.

The force needed to keep an object moving in a circle depends on the speed of the object and the radius of the circle. The greater the speed and/or the smaller the radius, the greater the force needed. If a satellite or planet slows down, it will be pulled in to a smaller radius orbit.

The solar system was formed over long periods from clouds of gases and dust drawn together by the force of gravity. When a force is used to compress a gas, work is done on the gas, leading to an increase in temperature.

During the formation of a star such as the Sun, a cloud of gas is pulled together by gravity, its temperature increases and the hydrogen nuclei gain sufficient energy to fuse into helium nuclei, releasing more energy.

The Universe contains thousands of millions of galaxies. The light coming from distant galaxies shows a red-shift that suggests that distant galaxies are moving away from us. The further away a galaxy is, the faster it is moving away from us; this suggests that space itself is expanding. Scientists' explanation for these observations is that the Universe began with a 'Big Bang' about 14 thousand million years ago.

The acceptance of the 'Big Bang' model to describe the early stages of the Universe depends on the interpretation of observations, as more observations were made, the theory became more secure.

Telescope designs have improved over the last 100 years, and modifications have made it possible to observe regions of the electromagnetic spectrum other than visible light. Placing these instruments outside the atmosphere has improved the range and quality of data obtained, and these improved data have increased the confidence in the 'Big Bang' model (IaS3).

Assessable learning outcomes
Learners will be able to:

1.	recall the main features of our solar system, including the similarities and distinctions between the planets, their moons, and artificial satellites
2.	**explain, for the circular orbits, how the force of gravity can lead to changing velocity of a planet but unchanged speed**
3.	**explain how, for a stable orbit, the radius must change if this speed changes** *qualitative only*
4.	recall that the solar system was formed from dust and gas drawn together by gravity
5.	**use the particle model of matter to explain how doing work on a gas can increase its temperature (e.g. bicycle pump, in stars)**
6.	explain how the Sun was formed when collapsing cloud of dust and gas resulted in fusion reactions, leading to an equilibrium between gravitational collapse and expansion due to the fusion energy
7.	explain the red-shift of light from galaxies which are receding *qualitative only*
8.	explain that the relationship between the distance of each galaxy and its speed is evidence of an expanding universe model
9.	explain how the evidence of an expanding universe leads to the 'Big Bang' model

Linked learning opportunities

Specification links
- **Nuclear fusion (P5.3).**

Practical work
- Investigate the relationship between the force, speed and radius of path for an object moving in a circle.

Ideas about Science
- Use the development of the 'Big Bang' model of the beginning of the Universe as an example of how scientific explanations become accepted (IaS3).